

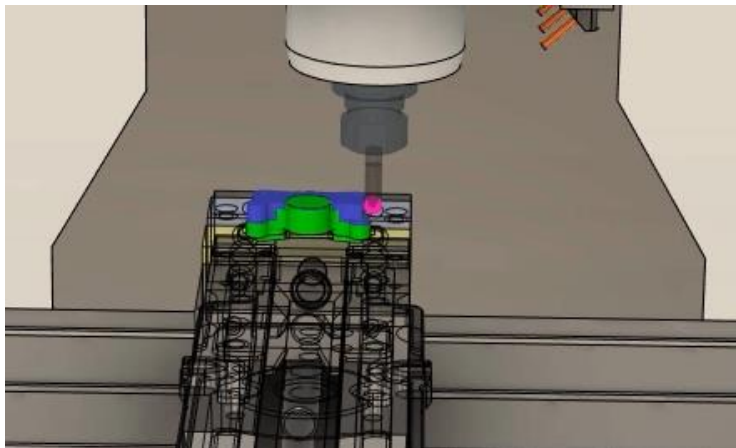
Step-by-step guide

Machine simulation

Import a machine model, locate a part on its table, then analyze a simulation from the setup.

Learning objectives:

- Review and run a Machine Simulation.



The completed exercise

1. Continue with the *Precision Machining Caliper – Front – Setup.f3z* file from the previous example.

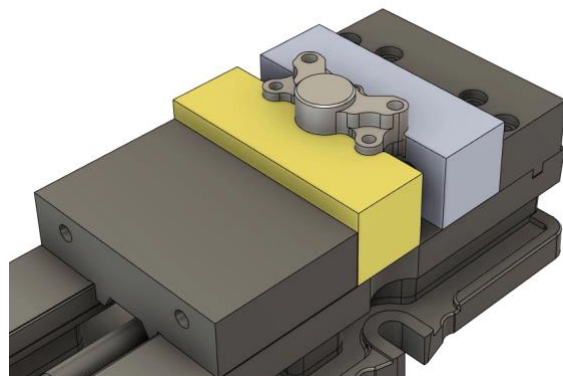


Figure 1. Continue with the file from the previous example

2. Simulating the part on a digital CNC machine can be used as a final check before outputting your code to a real CNC machine. Click Manage> Machine Library.

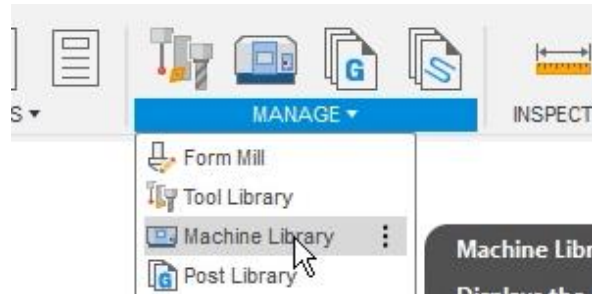


Figure 2. Open the machine library

3. Choose the Fusion 360 Library from the Machine Library dialog's left column.

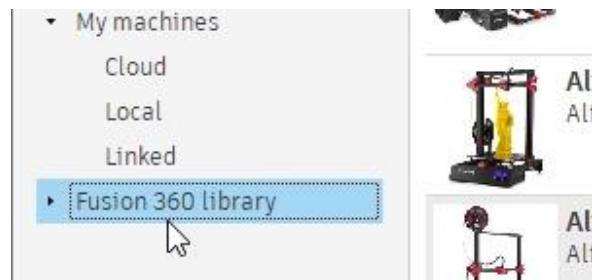


Figure 3. Open the Fusion 360 library

4. Filter the machines to show only HAAS simulation-ready milling machines.

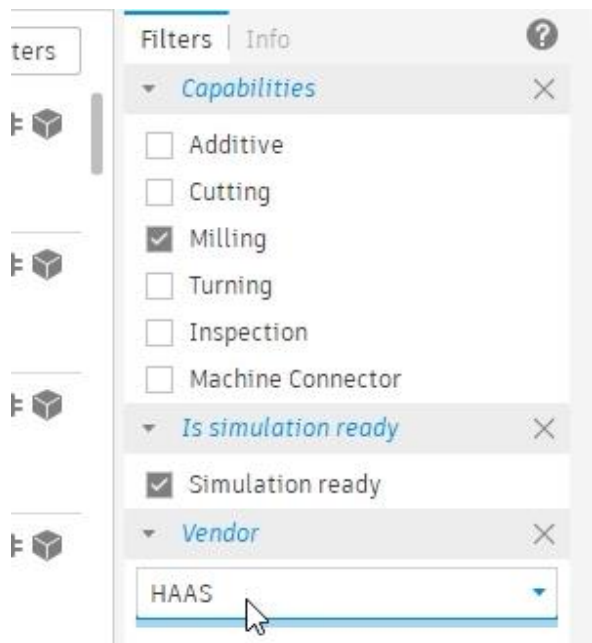


Figure 4. Filter the machines

5. Locate and select the HAAS VF-2 option without a rotary table.



Figure 5. Select a machine

6. Note the machine specifications listed in the dialog Info tab. Close the Machine Library dialog after you finish exploring the machine. Now that you know the machine is available in the specs are appropriate, it can be added to an existing setup.

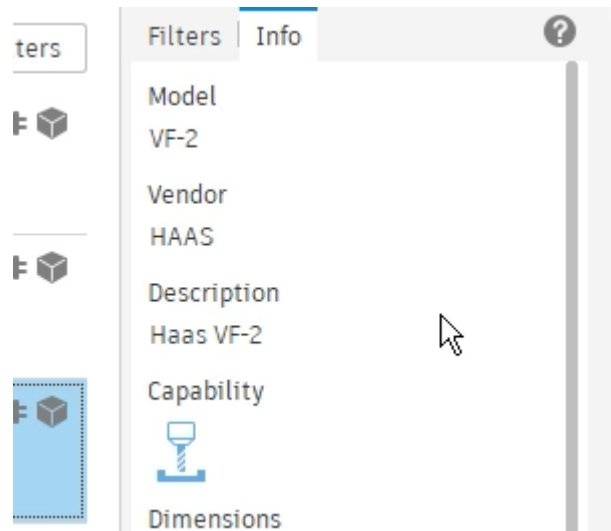


Figure 6. Inspect the machine's specs

7. Edit the OP2 setup.

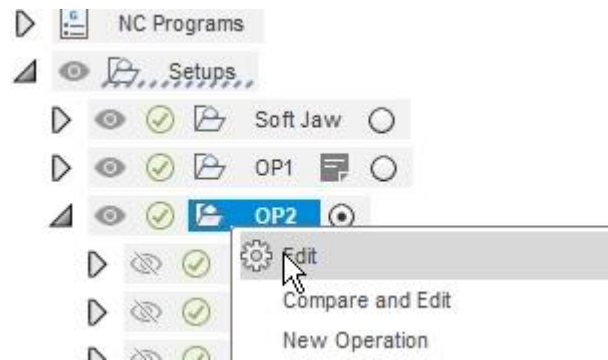


Figure 7. Edit the OP2 setup

8. Click the dialog's Select to choose the setup's machine.

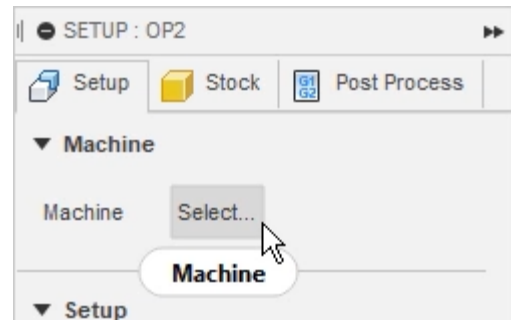


Figure 8. Click Select

9. Locate and select the same HAAS VF-2 machine you found in Step 5. Click the Machine Library dialog's Select.



Figure 9. Select the VF-2 machine

10. Click the dialog Download Model.



Figure 10. Click Download Model

11. Choose the machine file name and save location, then click the dialog Save.

Name:

haas vf-2.f3d

Location:

22 Precision Machining Caliper > S4 Update > M

Figure 11. Save the machine's file

12. Click the dialog No to avoid installing the machine connector.

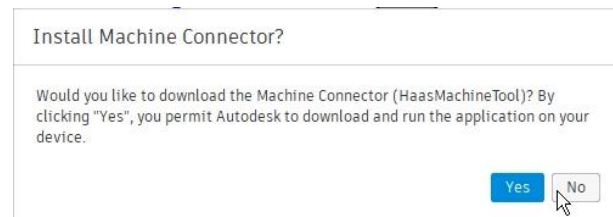


Figure 12. Click No

13. Navigate to the Setup dialog new Part Position tab and notice that you can position the fixture inside the machine.

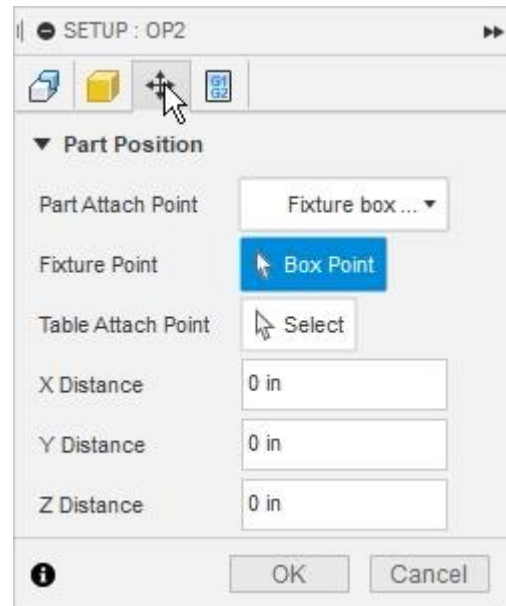


Figure 13. Navigate to the Part Position tab

14. Adjust the Y Distance value until the vise mounting holes align with the table T slot feature. OK the dialog.

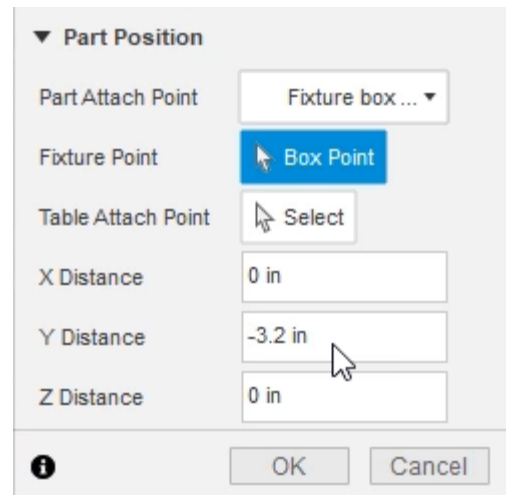


Figure 14. Align the vise to the machine

15. Inspect the machine to make sure it looks correct.

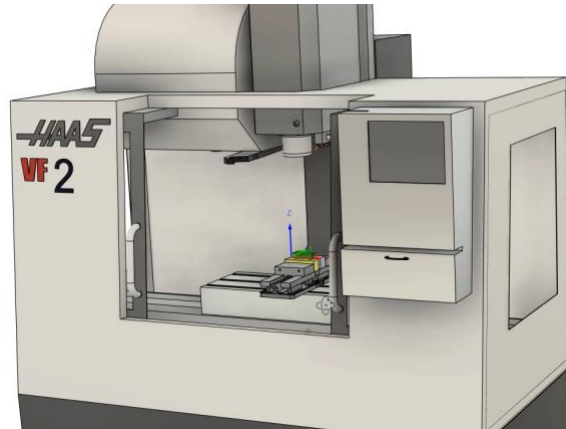


Figure 15. Inspect the machine

16. Select and activate the OP2 setup, then click Actions> Simulate with Machine.

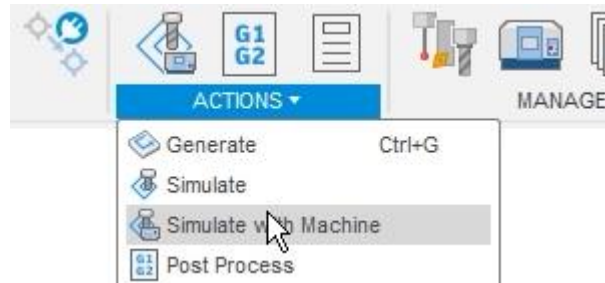


Figure 16. Simulate a setup with the machine

17. Press play and watch the animation. The simulations you watched in the past move the tool in relation to the part. When simulating with a machine, the part moves in relation to the tool which is the realistic way a part is machined. Exit the simulation after you finish analyzing it.

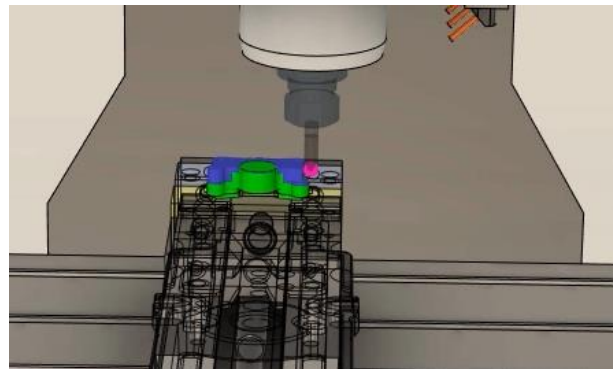


Figure 17. Watch the simulation

18. Click Show Data Panel to open the Data Panel.



Figure 18. Click Show Data Panel

19. Open the project Machine Data folder.

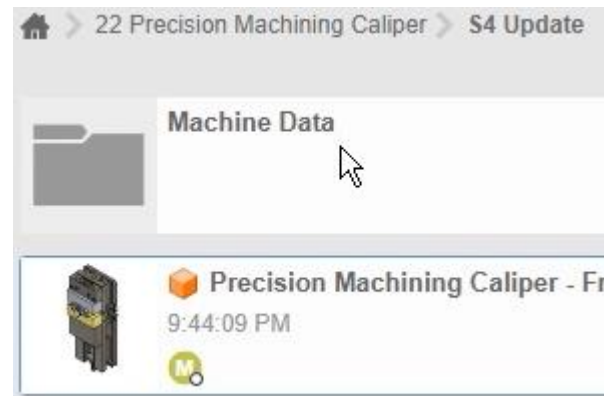


Figure 19. Open the Machine Data folder

20. Notice that the HAAS VF-2 machine is saved into this folder. Save the file.

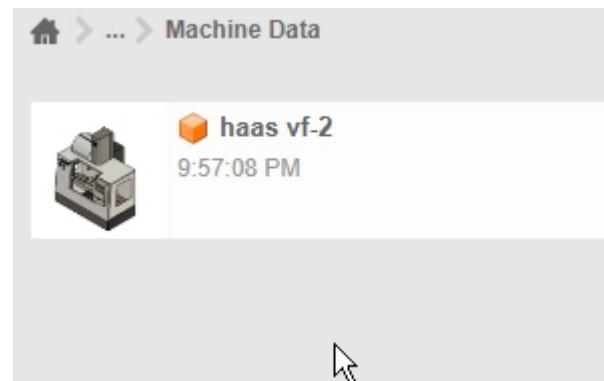


Figure 20. Notice the machine's file